# Functional Analysis

Problem

# Contents

**Functional Analysis: Introduction to Further Topics in Analysis** **1**
Chapter 1. $L^p$ Spaces and Banach Spaces . . . . . 1
Chapter 2. $L^p$ Spaces in Harmonic Analysis . . . . . 10
Chapter 3. Distributions: Generalized Functions . . . . . 20
Chapter 4. Applications of the Baire Category Theorem . . . . . 28
Chapter 5. Rudiments of Probability Theory . . . . . 35
Chapter 6. An Introduction to Brownian Motion . . . . . 40
Chapter 7. A Glimpse into Several Complex Variables . . . . . 47
Chapter 8. Oscillatory Integrals in Fourier Analysis . . . . . 53

# Functional Analysis: Introduction to Further Topics in Analysis

## Chapter 1. $L^p$ Spaces and Banach Spaces

**1.** The spaces $L^\infty$ and $L^1$ play universal roles with respect to all Banach spaces in the following sense.

(a) If $\mathcal{B}$ is any separable Banach space, show that it can be realized without change of norm as a linear subspace of $L^\infty(\mathbb{Z})$. Precisely, prove that there is a linear operator $i$ of $\mathcal{B}$ into $L^\infty(\mathbb{Z})$ so that $\|i(f)\|_{L^\infty(\mathbb{Z})} = \|f\|_{\mathcal{B}}$ for all $f \in \mathcal{B}$.

(b) Each such $\mathcal{B}$ can also be realized as a quotient space of $L^1(\mathbb{Z})$. That is, there is a linear surjection $P$ of $L^1(\mathbb{Z})$ onto $\mathcal{B}$, so that if $\mathcal{S} = \{x \in L^1(\mathbb{Z}) : \ P(x) = 0\}$, then $\|P(x)\|_{\mathcal{B}} = \inf_{y\in\mathcal{S}} \|x + y\|_{L^1(\mathbb{Z})}$, for each $x \in L^1(\mathbb{Z})$. This gives an identification of $\mathcal{B}$ (and its norm) with the quotient space $L^1(\mathbb{Z})/\mathcal{S}$ (and its norm), as defined in Exercise 30.

Note that similar conclusions hold for $L^\infty(X)$ and $L^1(X)$ if $X$ is a measure space that contains a countable disjoint collection of measurable sets of positive and finite measure.

[Hint: For (a), let $\{f_n\}$ be a dense set of non-zero vectors in $\mathcal{B}$, and let $\ell_n \in \mathcal{B}^*$ be such that $\|\ell_n\|_{\mathcal{B}^*} = 1$ and $\ell_n(f_n) = \|f_n\|$. If $f \in \mathcal{B}$, set $i(f) = \{\ell_n(f)\}_{-\infty}^{\infty}$. For (b), if $x = \{x_n\} \in L^1(\mathbb{Z})$, with $\sum_{-\infty}^{\infty} |x_n| = \|x\|_{L^1(\mathbb{Z})} < \infty$, define $P$ by $P(x) = \sum_{-\infty}^{\infty} x_n f_n/\|f_n\|$.]

**2.** There is a "generalized limit" $L$ defined on the vector space $V$ of all real sequences $\{s_n\}_{n=1}^{\infty}$ that are bounded, so that:

(i) $L$ is a linear functional on $V$.

(ii) $L(\{s_n\}) \geq 0$ if $s_n \geq 0$, for all $n$.

(iii) $L(\{s_n\}) = \lim_{n\to\infty} s_n$ if the sequence $\{s_n\}$ has a limit.

(iii) $L(\{s_n\}) = L(\{s_{n+k}\})$ for every $k \geq 1$.

(iii) $L(\{s_n\}) = L(\{s_{n'}\})$ if $s_n - s'_n \neq 0$ for only finitely many $n$.

[Hint: Let $p(\{s_n\}) = \limsup_{n\to\infty} \left(\frac{s_1+\cdots+s_n}{n}\right)$, and extend the linear functional $L$ defined by $L(\{s_n\}) = \lim_{n\to\infty} s_n$, defined on the subspace consisting of sequences that have limits.]

**3.** Show that the closed unit ball in a Banach space $\mathcal{B}$ is compact (that is, if $f_n \in \mathcal{B}$, $\|f_n\| \leq 1$, then there is a subsequence that converges in the norm) if and only if $\mathcal{B}$ is finite dimensional.

[Hint: If $\mathcal{S}$ is a closed subspace of $\mathcal{B}$, then there exists $x \in \mathcal{B}$ with $\|x\| = 1$ and the distance between $x$ and $\mathcal{S}$ is greater than $1/2$.]

**4.** Suppose $X$ is a $\sigma$-compact measurable metric space, and $C_b(X)$ is separable, where $C_b(X)$ denotes the Banach space of bounded continuous functions on $X$ with the sup-norm.

(a) If $\{\mu_n\}_{n=1}^{\infty}$ is a bounded sequence in $M(X)$, then there exists a $\mu \in M(X)$ and a subsequence $\{\mu_{n_j}\}_{j=1}^{\infty}$, so that $\mu_{n_j}$ converges to $\mu$ in the following (weak$^*$) sense:

$$\int_X g(x)\, d\mu_{n_j}(x) \to \int_X g(x)\, d\mu(x), \qquad \text{for all } g \in C_b(X).$$

(b) Start with a $\mu_0 \in M(X)$ that is positive, and for each $f \in L^1(\mu_0)$ consider the mapping $f \mapsto f d\mu_0$. This mapping is an isometry of $L^1(\mu_0)$ to the subspace of $M(X)$ consisting of signed measures which are absolutely continuous with respect to $\mu_0$.

(c) Hence if $\{f_n\}$ is a bounded sequence of functions in $L^1(\mu_0)$, then there exist a $\mu \in M(X)$ and a subsequence $\{f_{n_j}\}$ such that the measures $f_{n_j} d\mu_0$ converge to $\mu$ in the above sense.

**5.*** Let $X$ be a measure space. Suppose $\varphi$ and $\psi$ are both continuous, strictly increasing functions on $[0, \infty)$ which are inverses of each other, that is, $(\varphi \circ \psi)(x) = x$ for all $x \geq 0$. Let

$$\Phi(x) = \int_0^x \varphi(u)\, du \qquad \text{and} \qquad \Psi(x) = \int_0^x \psi(u)\, du.$$

Consider the Orlicz spaces $L^\Phi(X)$ and $L^\Psi(X)$ introduced in Exercise 23.

(a) In connection with Exercise 22 the following Hölder-like inequality holds:

$$\int |fg| \leq C \|f\|_{L^\Phi} \|g\|_{L^\Psi} \qquad \text{for some } C > 0, \text{ and all } f \in L^\Phi \text{ and } g \in L^\Psi.$$

(b) Suppose there exists $c > 0$ so that $\Phi(2t) \leq c\Phi(t)$ for all $t \geq 0$. Then the dual of $L^\Phi$ is equivalent to $L^\Psi$.

**6.*** There are generalizations of the parallelogram law for $L^2$ (see Exercise 25) that hold for $L^p$. These are the Clarkson inequalities:

(a) For $2 \leq p \leq \infty$ the statement is that

$$\left\| \frac{f+g}{2} \right\|_{L^p}^p + \left\| \frac{f-g}{2} \right\|_{L^p}^p \leq \frac{1}{2} \left( \|f\|_{L^p}^p + \|g\|_{L^p}^p \right).$$

(b) For $1 < p \leq 2$ the statement is that

$$\left\| \frac{f+g}{2} \right\|_{L^p}^q + \left\| \frac{f-g}{2} \right\|_{L^p}^q \leq \frac{1}{2} \left( \|f\|_{L^p}^p + \|g\|_{L^p}^p \right)^{q/p},$$

where $1/p + 1/q = 1$.

(c) As a result, $L^p$ is **uniformly convex** when $1 < p < \infty$. This means that there is a function $\delta = \delta(\epsilon) = \delta_p(\epsilon)$, with $0 < \delta < 1$, (and $\delta(\epsilon) \to 0$ as $\epsilon \to 0$), so that whenever $\|f\|_{L^p} = \|g\|_{L^p} = 1$, then $\|f - g\|_{L^p} \geq \epsilon$ implies that $\left\| \frac{f+g}{2} \right\| \leq 1 - \delta$.

This is stronger than the conclusion of strict convexity in Exercise 27.

(d) Using the result in (c), prove the following: suppose $1 < p < \infty$, and the sequence $\{f_n\}$, $f_n \in L^p$, converges weakly to $f$. If $\|f_n\|_{L^p} \to \|f\|_{L^p}$, then $f_n$ converges to $f$ strongly, that is, $\|f_n - f\|_{L^p} \to 0$ as $n \to \infty$.

**7.**$^*$ An important notion is that of the equivalence of Banach spaces. Suppose $\mathcal{B}_1$ and $\mathcal{B}_2$ are a pair of Banach spaces. We say that $\mathcal{B}_1$ and $\mathcal{B}_2$ are **equivalent** (also said to be "isomorphic") if there is a linear bijection $T$ between $\mathcal{B}_1$ and $\mathcal{B}_2$ that is bounded and whose inverse is also bounded. Note that any pair of finite-dimensional Banach spaces are equivalent if and only if their dimensions are the same.

Suppose now we consider $L^p(X)$ for a general class of $X$ (which contains for instance, $X = \mathbb{R}^d$ with Lebesgue measure). Then:

(a) $L^p$ and $L^q$ are equivalent if and only if $p = q$.

(b) However, for any $p$ with $1 \leq p \leq \infty$, $L^2$ is equivalent with a closed infinite-dimensional subspace of $L^p$.

**8.**$^*$ There is no finitely-additive rotationally-invariant measure extending Lebesgue measure to all subsets of the sphere $S^d$ when $d \geq 2$, in distinction to what happens on the torus $\mathbb{R}^d/\mathbb{Z}^d$ when $d \geq 2$. (See Exercise 36). This is due to a remarkable construction of Hausdorff that uses the fact that the corresponding rotation group of $S^d$ is non-commutative. In fact, one can decompose $S^2$ into four disjoint sets $A$, $B$, $C$ and $Z$ so that (i) $Z$ is denumerable, (ii) $A \sim B \sim C$, but $A \sim (B \cup C)$.

Here the notation $A_1 \sim A_2$ means that $A_1$ can be transformed into $A_2$ via a rotation.

**9.**$^*$ As a consequence of the previous problem one can show that it is not possible to extend Lebesgue measure on $\mathbb{R}^d$, $d \geq 3$, as a finitely-additive measure on all subsets of $\mathbb{R}^d$ so that it is both translation and rotation invariant (that is, invariant under Euclidean motions). This is graphically shown by the "Banach-Tarski paradox": There is a finite decomposition of the unit ball $B_1 = \bigcup_{j=1}^N E_j$, with the sets $E_j$ disjoint, and there are corresponding sets $\tilde{E}_j$ that are each obtained from $E_j$ by a Euclidean motion, with the $\tilde{E}_j$ also disjoint, so that $\bigcup_{j=1}^N \tilde{E}_j = B_2$ the ball of radius 2.

## Chapter 2. $L^p$ Spaces in Harmonic Analysis

**1.** Another way $\mathbf{H}^1_r$ is an improvement over $L^1$ is in its weak compactness of the unit ball. The following can be proved. Suppose $\{f_n\}$ is a sequence in $\mathbf{H}^1_r$ with $\|f_n\|_{\mathbf{H}^1_r} \leq A$. Then we can select a subsequence $\{f_{n_k}\}$ and find an $f \in \mathbf{H}^1_r$ so that $\int f_{n_k}(x)\varphi(x)\,dx \to \int f(x)\varphi(x)\,dx$, as $k \to \infty$, for every $\varphi$ that is a continuous function of compact support.

This is to be compared with $L^1$, and the failure there of weak compactness as described in Exercises 12 and 13 in the previous chapter.

[Hint: Apply the result in Problem 4 (c) of the previous chapter to obtain a subsequence $\{f_{n_k}\}$ and a finite measure $\mu$ so that $f_{n_k} \to \mu$ in the weak$^*$ sense. Next use the fact that if $\sup_{\epsilon>0} |\mu * \varphi_\epsilon| \in L^1$, for an appropriate $\varphi$, then $\mu$ is absolutely continuous.]

**2.** Suppose $H^p$ is the complex Hardy space defined in Section 5. For $1 < p < \infty$, prove the following:

(a) If $F \in H^p$, then $\lim_{y\to 0} F(x+iy) = F_0(x)$ exists in the $L^p(\mathbb{R})$ norm.

(b) $\|F\|_{H^p} = \|F_0\|_{L^p}$.

(c) One has $2F_0 = f + iH(f)$, with $f$ real-valued in $L^p(\mathbb{R})$, and $\|F_0\|_{L^p} \approx \|f\|_{L^p}$. Moreover, every $F_0$ (and thus $F$) arises this way. This gives a linear isomorphism (over the reals) of $H^p$, with $L^p$ with an equivalence of norms.

[Hint: Here is an outline of the proof. For each $y_1 > 0$, write $F_{y_1}(z) = F(z + iy_1)$ and $F^\epsilon_{y_1}(z) = F_{y_1}(z)/(1 - i\epsilon z)$, $\epsilon > 0$. One has that $F_{y_1}$ is bounded $\overline{\mathbb{R}^2_+}$ (see Section 2, in Chapter 5, Book III). Thus by Exercise 9, $F^\epsilon_{y_1}(z) = (F^\epsilon_{y_1} * \mathcal{P}_y)(x)$. Now using the weak compactness of the unit ball in $L^p$, (Exercise 12 in Chapter 1), we can find $F_0 \in L^p$ so that $F^\epsilon_{y_1}(x) \to F_0(x)$ weakly as $\epsilon$ and $y_1 \to 0$. Observe that this breaks down for $p = 1$. Conclusion (c) is then essentially a restatement of the boundedness of the Hilbert transform for $1 < p < \infty$.]

**3.** Let $P$ be any non-zero polynomial of degree $k$ in $\mathbb{R}^d$. Then $f = \log|P(x)|$ is in BMO and $\|f\|_{\mathrm{BMO}} \leq c_k$, where $c_k$ depends only on the degree $k$ of the polynomial.

[Hint: Verify the result first when $d = 1$. Then use induction in the dimension and the following assertion, stated for $\mathbb{R}^2$. Suppose $f(x, y)$, $(x, y) \in \mathbb{R}^2$ is for each $y$ a $\mathrm{BMO}(\mathbb{R})$ function in $x$, uniformly in $y$. Assume also that this holds when the roles of $x$ and $y$ are interchanged. Then $f \in \mathrm{BMO}(\mathbb{R}^2)$.]

**4.** Prove the following John-Nirenberg inequalities for every $f \in \mathrm{BMO}(\mathbb{R}^d)$:

(a) For every $q < \infty$ there is a bound $b_q$ so that

$$\sup_B \frac{1}{m(B)} \int_B |f - f_B|^q \, dx \leq b_q^q \|f\|_{\mathrm{BMO}}^q.$$

(b) There are positive constants $\mu$ and $A$, so that

$$\sup_B \frac{1}{m(B)} \int_B e^{\mu|f-f_B|} \, dx \leq A, \qquad \text{whenever } \|f\|_{\mathrm{BMO}} \leq 1.$$

[Hint: For (a) test $f$ against $p$-atoms, where $p$ is dual to $q$. For (b) use the bound $c_p = O(1/(p-1))$ as $p \to 1$ (in (38)) to obtain $b_q = O(q)$, as $q \to \infty$. Then write $e^u = \sum_{q=0}^{\infty} u^q/q!$.]

**5.** The Hilbert transform of a bounded function is in BMO. Show this in two different ways.

(a) Directly: Suppose $f$ is bounded (and belongs to some $L^p$, $1 \leq p < \infty$). Then $H(f) \in \text{BMO}$ with

$$\|H(f)\|_{\text{BMO}} \leq A\|f\|_{L^\infty},$$

with $A$ not depending on the $L^p$ norm of $f$.

(b) By duality, using Theorem 5.4.

[Hint: For (a), fix any ball $B$, and let $B_1$ be its double. Consider separately, $f\chi_{B_1}$ and $f\chi_{B_1^c}$.]

**6.*** The following is the maximal characterization of $\mathbf{H}^1_r(\mathbb{R}^d)$. Suppose $\Phi$ belongs to the Schwartz space $\mathcal{S}$ and $\int \Phi(x)\,dx \neq 0$. Let $M(f)(x) = \sup_{\epsilon>0} |(f * \Phi_\epsilon)(x)|$, for $f \in L^1$. Then

(a) $f$ is in $\mathbf{H}^1_r$ if and only if $M(f)$ belongs to $L^1$.

(b) The condition $\Phi \in \mathcal{S}$ can be relaxed to require only

$$|\partial_x^\alpha \Phi(x)| \leq c_\alpha (1+|x|)^{-d-1-|\alpha|}.$$

(c) Note two interesting examples, first $\Phi_{t^{1/2}}(x) = (4\pi t)^{-d/2} e^{-|x|^2/(4t)}$: then $u(x,t) = (f * \Phi_{t^{1/2}})(x)$ is the solution of the Heat equation $\triangle_x u = \partial_t u$, with initial data $u(x,0) = f(x)$. Also, $\Phi_t(x) = \frac{c_d t}{(t^2+|x|^2)^{\frac{d+1}{2}}}$ with $c_d = \Gamma(\frac{d+1}{2})/\pi^{\frac{d+1}{2}}$ so that $u(x,t) = (f * \Phi_t)(x)$ is the solution to Laplace's equation $\triangle_x u + \partial_t^2 u = 0$, with initial data $u(x,0) = f(x)$. (Here $\Gamma$ denotes the gamma function.)

**7.*** $H^p$, when $p = 1$. The results in (a) and (b) of Problem 2 also hold for $p = 1$, but require a different proof. The analog of (c) is as follows. One has $2F_0 = f + iH(f)$, where $f$ belongs to the real Hardy space $\mathbf{H}^1_r$. Also $\|F_0\|_{L^1} \approx \|f\|_{\mathbf{H}^1_r}$. As a consequence, a necessary and sufficient condition that $f \in \mathbf{H}^1_r$ is that both $f$ and $H(f)$ are in $L^1$.

The conclusions (a) and (b) may be proved by showing that any $F \in H^1$ can be written as $F = F_1 \cdot F_2$ with $F_j \in H^2$ and $\|F_j\|^2_{H^2} = \|F\|_{H^1}$, and then using the corresponding results in $H^2$.

**8.*** Suppose $f \in L^1(\mathbb{R})$. Then we can define $H(f) \in L^1(\mathbb{R})$ in the weak sense to mean that there exists $g \in L^1(\mathbb{R})$ so that

$$\int_{\mathbb{R}} g\varphi \, dx = \int_{\mathbb{R}} fH(\varphi) \, dx, \qquad \text{for all functions } \varphi \text{ in the Schwartz space.}$$

Then we say $g = H(f)$ in the weak sense.

As a consequence of Problem 7*, one has that $f \in \mathbf{H}^1_r(\mathbb{R})$ if and only if $f \in L^1(\mathbb{R})$ and $H(f)$, taken in the weak sense, also belongs to $L^1(\mathbb{R})$.

**9.*** Let $\{f_n\}$ be a sequence of elements in $\mathbf{H}^1_r$ so that $\|f_n\|_{\mathbf{H}^1_r} \leq M < \infty$ for all $n$. Assume that $f_n$ converges to $f$ almost everywhere. Then:

(a) $f \in \mathbf{H}^1_r$.

(b) $\int f_n g \to \int fg$, as $n \to \infty$, for all $g$ continuous with compact support.

A corresponding result holds for $L^p$, $p > 1$, but fails for $p = 1$. See Exercise 14 in Chapter 1.

**10.*** The following result illustrates the application of $\mathbf{H}^1_r$ to the theory of compensated compactness.

Suppose $A = (A_1, \ldots, A_d)$ and $B = (B_1, \ldots, B_d)$ are vector fields in $\mathbb{R}^d$ with $A_i, B_i \in L^2(\mathbb{R}^d)$ for all $i$. The divergence of $A$ is defined by

$$\operatorname{div}(A) = \sum_{k=1}^{d} \frac{\partial A_k}{\partial_{x_k}},$$

and the curl of $B$ is the $d \times d$ matrix whose $ij$-entry is

$$(\operatorname{curl}(B))_{ij} = \frac{\partial B_i}{\partial x_j} - \frac{\partial B_j}{\partial x_i}.$$

(The derivatives here are taken in the sense of distributions, as in the next chapter.)
If $\operatorname{div}(A) = 0$ and $\operatorname{curl}(B) = 0$ then $\sum_{k=1}^{d} A_k B_k \in \mathbf{H}^1_r$.
This is in contrast with the result that in general, if $f, g \in L^2$, then one only has $fg \in L^1$.

## Chapter 3. Distributions: Generalized Functions

**1.** We consider **periodic distributions** and their Fourier series.

(a) The notion of a periodic distribution on $\mathbb{R}^d$ can be defined in two equivalent ways:

First, one can consider distributions $F$ on $\mathbb{R}^d$ which are periodic in the sense that $\tau_h(F) = F$ for all $h \in \mathbb{Z}^d$;

Alternatively, one can consider the continuous linear functionals on $\mathcal{D}(\mathbb{T}^d)$, the space of $C^\infty$ periodic functions on $\mathbb{R}^d$. (Here $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ denotes the $d$-dimensional torus.)

(b) Note that if $\varphi \in \mathcal{D}(\mathbb{T}^d)$, then $\varphi$ has a Fourier series expansion

$$\varphi(x) = \sum_n a_n e^{2\pi i n\cdot x},$$

where the Fourier coefficients $a_n = \int_{\mathbb{T}^d} f(x)e^{-2\pi i n\cdot x}\,dx$ are rapidly decreasing, that is, for every $N > 0$, $|a_n| \le O(|n|^{-N})$ as $|n| \to \infty$.

Similarly, if $F$ is a periodic distribution, and $a_n = F(e^{-2\pi i n x})$ denote its Fourier coefficients, then $a_n$ are slowly increasing in the sense that for some $N > 0$, $|a_n| \le O(|n|^N)$ as $|n| \to \infty$.

Moreover, the Fourier series $\sum a_n e^{2\pi i n x}$ converges to $F$ in the sense of distributions.

[Hint: To prove the equivalence in (a), consider the "periodization" operator $P : \mathcal{D}(\mathbb{R}^d) \to \mathcal{D}(\mathbb{T}^d)$,

$$P(\varphi)(x) = \sum_{h\in\mathbb{Z}^d} \tau_h(\varphi)(x) = \sum_{h\in\mathbb{Z}^d} \varphi(x-h).$$

Then find $\gamma \in \mathcal{D}(\mathbb{R}^d)$ so that $P(\gamma) = 1$. This allows to prove that $P$ is surjective, and that, in the same way, its dual $P^* : \mathcal{D}_2^* \to \mathcal{D}_1^*$ is also surjective. (Here $\mathcal{D}_1^*$ and $\mathcal{D}_2^*$ denote, respectively, the two spaces of distributions described in (a).) To construct $\gamma$, pick $\psi \in \mathcal{D}(\mathbb{R}^d)$ so that $\psi \ge 0$ and $\psi = 1$ on $\{0 \le x_j < 1,\ \ 1 \le j \le d\}$, and let $\gamma = \psi/P(\psi)$.]

**2.** Suppose $Tf = f * K$ is a singular integral operator as in Theorem 3.2 of Section 3. Then the mapping $f \mapsto T(f)$ is bounded on the Hardy space $\mathbf{H}^1_r$, and in particular maps $\mathbf{H}^1_r$ to $L^1$.

[Hint: Consider first a 2-atom $\mathfrak{a}$ associated to the unit ball $B$. Then for an appropriate constant $c$ (bounded independently of $\mathfrak{a}$) we have $T(\mathfrak{a}) = c(a_* + \Phi)$. Here $a_*$ is a 2-atom for the ball $B^a st$, the double of $B$, and $\Phi$ satisfies $|\Phi(x)| \leq (1 + |x|)^{-d-1}$, $\int_{\mathbb{R}^d} \Phi(x)\, dx = 0$. With this apply Exercise 21 in Chapter 2. Then obtain the analog for 2-atoms $\mathfrak{a}$, after rescaling and translation.]

**3.** Prove the following interior estimates for an elliptic operator $L$ of order $m$ with constant coefficients.

Suppose $\mathcal{O}$ and $\mathcal{O}_1$ are bounded subsets of $\mathbb{R}^d$ with $\overline{\mathcal{O}} \subset \mathcal{O}_1$. Assume $u$ and $f$ are $L^p$ functions in $\mathcal{O}_1$ with $Lu = f$ in $\mathcal{O}_1$ in the sense of distributions. Then if $1 < p < \infty$ and $k$ is a non-negative integer, we have

$$\sum_{|\alpha| \leq m+k} \|\partial_x^\alpha u\|_{L^p(\mathcal{O})} \leq c \left( \sum_{|\beta| \leq k} \|\partial_x^\beta f\|_{L^p(\mathcal{O}_1)} + \|u\|_{L^p(\mathcal{O}_1)} \right)$$

where the derivatives are taken in the sense of distributions.

[Hint: Consider the parametrix $Q_\epsilon = \eta_\epsilon Q$ given in Corollary 2.13 which is supported in $|x| \leq \epsilon$. Here $\epsilon$ is chosen so that $\overline{\mathcal{O}}_\epsilon \subset \mathcal{O}_1$, where $\mathcal{O}_\epsilon$ are the points of distance $\leq \epsilon$ from $\mathcal{O}$.

Set $U = \psi u$, with $\psi$ a $C^\infty$ function that is 1 near $\overline{\mathcal{O}}_\epsilon$ but vanishes outside $\mathcal{O}_1$. Then

$$L(U) = \psi L(u) + \sum_{|\gamma| < m} \psi_\gamma \partial_x^\gamma u,$$

and what is important is that the $\psi_\gamma$ vanishes in $\overline{\mathcal{O}}_\epsilon$. Now $U + r_\epsilon * U = Q_\epsilon * L(U)$, where $r_\epsilon \in \mathcal{S}$. This gives

$$\psi u = Q_\epsilon * (\psi f) - r_\epsilon(\psi u) + \sum_r Q_\epsilon * (\psi_\gamma \partial_x^\gamma u).$$

As has been pointed out, $\partial_x^\alpha Q$ are Calderón-Zygmund distributions whenever $|\gamma| \leq m$, so the same is true for $Q_\epsilon$. Then using Theorem 3.2, the result follows.]

**4.*** Let $P(x)$ be any real polynomial in $\mathbb{R}^d$, and $k$ a homogeneous function of degree $-d$ with $\int_{|x|=1} k(x)\, d\sigma(x) = 0$.

(a) One can define the tempered distribution $\mathrm{pv}\left(e^{iP(x)}k(x)\right) = K$ by

$$K(\varphi) = \lim_{\epsilon \to 0} \int_{|x|\geq \epsilon} e^{iP(x)} k(x)\varphi(x)\, dx.$$

(b) Then the Fourier transform of $K$ is a bounded function (with bound independent of the coefficients of $P$).

**5.**$^*$ Let $Q$ be a fixed real-valued polynomial on $\mathbb{R}^d$. Consider the distributions initially defined for $\mathrm{Re}(s) > 0$ by

$$I(s)(\varphi) = \int_{Q(x)>0} |Q(x)|^s \varphi(x)\, dx, \qquad \text{where } \varphi \in \mathcal{S}.$$

Then $I(s)(\varphi)$ has a meromorphic continuation to the whole complex $s$-plane, with poles at most at $s = -k/m$, where $m$ is a positive integer determined by $Q$, and $k$ is any positive integer. The order of the poles do not exceed $d$.

**6.*** As a consequence of the results in Problem 5*, one may prove the following.

(a) Suppose $L = \sum_{|\alpha| \leq m} a_\alpha \partial_x^\alpha$ is a non-zero partial differential operator on $\mathbb{R}^d$ with $a_\alpha$ complex constants. Then $L$ has a *tempered* fundamental solution. As an immediate corollary we also have:

(b) Suppose $P$ is a complex-valued polynomial on $\mathbb{R}^d$. Then there exists a tempered distribution $F$ that agrees with $1/P$ where $P(x) \neq 0$.

In fact, let $P$ be the characteristic polynomial of $L$ and apply the result of the previous problem to $Q = |P|^2$. Suppose $I(s)$ has a pole of order $r$ at $s = 1$, then define the tempered distribution $F$ by

$$F = \overline{P} \, \frac{1}{r!} \frac{d^r}{ds^r} (s+1)^r I(s) \Big|_{s=-1}.$$

Consequently, $PF = 1$, and the inverse Fourier transform of $F$ is the desired fundamental solution of $L$.

**7.*** Suppose $L = \sum_{|\alpha| \leq m} a_\alpha \partial_x^\alpha$ is a partial differential operator on $\mathbb{R}^d$, with $a_\alpha$ complex constants. Then $L$ is hypo-elliptic if and only if for each $\alpha \neq 0$

$$\frac{\partial_\xi^\alpha P(\xi)}{P(\xi)} \to 0 \qquad \text{as } |\xi| \to \infty,$$

where $P$ denotes the characteristic polynomial of $L$.

**8.*** We describe several fundamental solutions of the wave operator

$$\Box = \frac{\partial^2}{\partial t^2} - \triangle_x,$$

where $(x,t) \in \mathbb{R}^d \times \mathbb{R}$ and $\triangle_x = \sum_{j=1}^d \frac{\partial^2}{\partial x_j^2}$.

We let $\Gamma_+$ be the forward open cone $= \{(x,t) : \ t > |x|\}$, and $\Gamma_- = -\Gamma_+$, the backward cone. For each $s$ with $\mathrm{Re}(s) > -1$ we define the function $F_s$ by

$$F_s(x,t) = \begin{cases} a_s(t^2 - |x|^2)^{s/2}, & \text{if } (x,t) \in \Gamma_+ \\ 0 & \text{otherwise.} \end{cases} \tag{29}$$

Here $a_s^{-1} = 2^{s+d}\pi^{\frac{d-1}{2}}\Gamma\left(\frac{s+d+1}{2}\right)\Gamma\left(s/2+1\right)$. Then $s \mapsto F_s$ has an analytic continuation in the complex $s$ plane as an entire (tempered) distribution-valued function. Moreover, one can prove that $F_+ = F_s|_{s=-d+1}$ is a fundamental solution of $\Box$.

Note that $F_-$, obtained from $F_+$ by mapping $t \mapsto -t$, is also a fundamental solution, and $F_+$ and $F_-$ are supported in $\overline{\Gamma}_+$ and $\overline{\Gamma}_-$ respectively. In addition, if $d$ is odd and $d \geq 3$, then $a_s$ vanishes for $s = -d+1$, so both $F_+$ and $F_-$ are supported on the boundary of their cones, which is a reflection of the Huygens' principle.

Finally, a third fundamental solution $F_0$ of interest is given by

$$F_0^\wedge = \lim_{\epsilon \to 0,\, \epsilon > 0} \frac{1}{4\pi^2}\left(\frac{1}{|\xi|^2 - \tau^2 + i\epsilon}\right),$$

with the limit taken in the sense of distribution, and $(\xi, \tau)$ representing the dual variables to $(x,t)$. The fundamental solutions $F_+$, $F_-$, and $F_0$ are each homogeneous of degree $-2$, and invariant under the Lorentz group of linear transformations of determinant 1 that preserves $\Gamma_+$. Also each fundamental solution of $\Box$ with these invariance properties can be written as $c_1F_+ + c_2F_- + c_3F_0$, with $c_1 + c_2 + c_3 = 1$.

## Chapter 4. Applications of the Baire Category Theorem

**1.** Let $T : \mathcal{B}_1 \to \mathcal{B}_2$ be a bounded linear transformation of a Banach space $\mathcal{B}_1$ to a Banach space $\mathcal{B}_2$.

(a) Prove that either $T$ is surjective, or the image $T(\mathcal{B}_1)$ is of the first category in $\mathcal{B}_2$.

(b) As a consequence, prove the following: Suppose $(X, \mu)$ is a finite measure space, and $1 \leq p_1 < p_2 \leq \infty$. One has of course $L^{p_2}(X) \subset L^{p_1}(X)$. Show that $L^{p_2}(X)$ is a set of the first category in $L^{p_1}(X)$ (except in the trivial case for which each element of $L^{p_1}$ belongs to $L^{p_2}$).

[Hint: For (a), assume that $T(\mathcal{B}_1)$ is of the second category and use an argument similar to the proof of Theorem 3.1 to show that the image under $T$ of a ball centered at the origin of $\mathcal{B}_1$ contains a ball centered at the origin in $\mathcal{B}_2$.]

**2.** For each integer $n \geq 2$, let $\Lambda_n$ denote the set of real numbers $x$ so that there exists infinitely many distinct fractions $p/q$ so that

$$|x - p/q| \leq 1/q^n.$$

Show that:

(a) $\Lambda_n$ is a generic set in $\mathbb{R}$.

(b) However, the Hausdorff dimension of $\Lambda_n$ equals $2/n$.

(c) Hence $m(\Lambda_n) = 0$, if $n > 2$, where $m$ denotes the Lebesgue measure.

The elements of $\Lambda = \bigcap_{n \geq 2} \Lambda_n$ are called the **Liouville numbers**. While it is not difficult to see that every element of $\Lambda$ is transcendental, it is a deeper fact that the same holds for each element of $\Lambda_n$ when $n > 2$. (Note that in the case $n = 2$, the set $\Lambda$ consists of the irrationals.)

**3.** Consider the Banach space $\mathcal{B}$ of continuous functions on the circle (with the sup-norm). Prove that the set of $f$ in $\mathcal{B}$ whose Fourier series diverges in a generic set on the circle, is itself a generic set in $\mathcal{B}$.

[Hint: Choose $\{x_i\}$ dense in $[0,1]$, let $E_i = \{f \in \mathcal{B} : \sup_N |S_N(f)(x_i)| = \infty\}$, and $E = \cap E_i$. Then $E$ is generic. For each $f \in E$, define $\mathcal{O}_n = \{x : |S_N(f)(x)| > n \text{ some } N\}$. Show that $\cap \mathcal{O}_n$ is generic.]

**4.** Let $\mathbb{D}$ denote the open unit disc in the complex plane, and let $\mathcal{A}$ be the Banach space of all continuous complex-valued functions on $\overline{\mathbb{D}}$ that are holomorphic on $\mathbb{D}$,

equipped with the sup-norm. Then, the space of functions in $\mathcal{A}$ which cannot be extended analytically past any point of the boundary of $\mathbb{D}$ is generic. To prove this statement establish the following:

(a) The set $\mathcal{A}_N = \{f \in \mathcal{A} : \ |f(e^{i\theta}) - f(1)| \leq N|\theta|\}$ is closed.

(b) $\mathcal{A}_N$ is nowhere dense.

[Hint: For (b) use the function $f_0(z) = (1-z)^{1/2}$ and consider $f + \epsilon f_0$.]

**5.** Let $I = [0, 1]$ denote the unit interval, and $C^\infty(I)$ the vector space of all smooth functions on $I$ equipped with the metric $d$ given by

$$d(f, g) = \sum_{n=0}^{\infty} \frac{1}{2^n} \frac{\rho_n(f - g)}{1 + \rho_n(f - g)},$$

where $\rho_n(h) = \sup_{x \in I} |h^{(n)}(x)|$. A function $f \in C^\infty(I)$ is analytic at a point $x_0 \in I$, if its Taylor series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

converges in a neighborhood of $x_0$ to the function $f$. The function $f$ is said to be singular at $x_0$ if its Taylor series diverges at $x_0$.

(a) Show that $(C^\infty(I), d)$ is a complete metric space.

(b) Prove that the set of functions in $C^\infty(I)$ that are singular at every point is generic.

[Hint: For (b), consider the set $F_K$ of smooth functions $f$ that satisfy $|f^{(n)}(x^*)|/n! \leq K^n$ for some $x^*$ and all $n$, and show that $F_K$ is closed and nowhere dense.]

**6.** The space $L^\infty$ in Theorem 4.2 cannot be replaced by any $L^q$, with $1 \leq q < \infty$. In fact there exists a closed infinite dimensional subspace of $L^1([0,1])$ consisting of functions that belong to $L^q$ for all $1 \leq q < \infty$.

[Hint: One may use Exercise 19 in the next chapter.]

**7.*** As an application of Exercise 14, let $\mathcal{H}$ denote the vector space of entire functions, that is, the set of functions that are holomorphic in all of $\mathbb{C}$. Given a compact subset $K$ of the complex plane and $f \in \mathcal{H}$, let $\|f\|_K = \sup_{z\in K} |f(z)|$. If $K_n$ denotes the closed disc centered at the origin and of radius $n$, define

$$d(f,g) = \sum_{n=1}^{\infty} \frac{1}{2^n} \frac{\|f-g\|_{K_n}}{1+\|f-g\|_{K_n}} \qquad \text{whenever } f, g \in \mathcal{H}.$$

Then $d$ is a metric, and $\mathcal{H}$ is a complete metric space with respect to $d$. Also, $d(f_n, f) \to 0$ if and only if $f_n$ converges to $f$ uniformly on every compact subset of $\mathbb{C}$.

Birkhoff's theorem (Problem 5, Chapter 2, Book II) states that there exists an entire function $F$ so that the set $\{F(z+n)\}_{n=1}^{\infty}$ is dense in $\mathcal{H}$. Also, MacLane's theorem (see the end of the same problem in Book II) says that there is an entire function $G$ so that the set of its derivatives $\{G^{(n)}(z)\}_{n=1}^{\infty}$ is dense in $\mathcal{H}$.

By Exercise 14, the set of functions in $\mathcal{H}$ with either of these properties is generic in $\mathcal{H}$, hence the set of entire functions with *both* properties is also generic.

## Chapter 5. Rudiments of Probability Theory

**1.** In the context of Bernoulli trials with probabilities $0 < p, q < 1$, where $p + q = 1$, let $D : \mathbb{Z}_2^\infty \to [0, 1]$, be given by

$$D(x) = \sum_{n=1}^{\infty} x_n/2^n \qquad \text{if } x = (x_1, \ldots, x_n, \ldots).$$

Under this mapping the measure $m_p$ goes to the measure $\mu_p$ that can be written symbolically as a "Riesz product," $\mu_p = \prod_{n=1}^{\infty}(1 + (p - q)r_n(t))\, dt$. The meaning of this is as follows. For each $N$, let

$$F_N(t) = \int_0^t \prod_{n=1}^{N} (1 + (p - q)r_n(s))\, ds.$$

Then one can show that:

(a) Each $F_N$ is increasing on $[0, 1]$.

(b) $F_N(0) = 0$, $F_N(1) = 1$.

(c) $F_N$ converges uniformly to a function $F$, as $N \to \infty$.

(d) $\mu_p = dF$, in the sense that $\mu_p((a, b)) = F(b) - F(a)$.

(e) If $p \neq 1/2$, then $\mu_p$ is completely singular (that is $dF/dt = 0$ almost everywhere.)

[Hint: Show that if $I = (a, b)$ is a dyadic interval of length $2^{-n}$, $a = \ell/2^n$ and $b = (\ell + 1)/2^n$, and $N \geq n$, then

$$F_N(b) - F_N(a) = p^{n_0} q^{n_1},$$

where $n_0$ is the number of zeroes in the first $n$ terms of the dyadic expansion of $\ell/2^n$, and $n_1$ is the number of 1's, with $n_0 + n_1 = n$.]

**2.*** There is an analogy between the Walsh-Paley expansion (see Exercise 16) and the Fourier expansion, that is, between $\{w_n\}_{n=0}^{\infty}$ and $\{e^{in\theta}\}_{n=-\infty}^{\infty}$. In this analogy the Rademacher functions $r_k = w_{2^{k-1}}$ correspond to the lacunary frequencies $\{e^{i2^k\theta}\}_{k=0}^{\infty}$. In fact, the following is known:

(a) If $\sum_{k=0}^{\infty} c_k e^{i2^k\theta}$ is an $L^2([0,2\pi])$ function, then it belongs to $L^p$, for every $p < \infty$.

(b) If $\sum_{k=0}^{\infty} c_k e^{i2^k\theta}$ is the Fourier series of an integrable function, it belongs to $L^2$, and hence to $L^p$ for every $p < \infty$.

(c) This function belongs to $L^\infty$ if and only if $\sum_{k=0}^{\infty} |c_k| < \infty$.

(d) From (c) it follows that the conclusion (a) of Theorem 1.7 does not necessarily extend to $p = \infty$.

**3.** The following is a general form of the central limit theorem. Suppose $f_1, \ldots, f_n, \ldots$ are square integrable mutually independent functions on $X$, and assume for simplicity that each has mean equal to 0. Let $\mu_n$ be the distribution measure of $f_n$, and $\sigma_n^2$ the variance. Set $S_n^2 = \sum_{k=1}^n \sigma_n^2$. The critical assumption is that for every $\epsilon > 0$

$$\lim_{n\to\infty} \frac{1}{S_n^2} \sum_{k=1}^{n} \int_{|t|\geq \epsilon S_n} t^2 \, d\mu_n(t) = 0.$$

Under these conditions the distribution measures of $\frac{1}{S_n} \sum_{k=1}^n f_k$ converge weakly to the normal distribution $\nu$ with variance 1.

**4.*** Suppose $\{f_n\}$ are identically distributed, square integrable, mutually independent, have each mean 0 and variance 1. Let $s_n = \sum_{k=1}^{n} f_k$. Then for a.e. $x$

$$\limsup_{n\to\infty} \frac{s_n(x)}{(2n \log\log n)^{1/2}} = 1.$$

This is the law of the "iterated logarithm."

**5.** An interesting variant of the random walk in $\mathbb{R}^d$ (often referred to as a "random flight") arises if the motion of unit distance at the $n^{\text{th}}$ step is allowed to be in *any* direction (of the unit sphere). More precisely

$$s_n = f_1 + \cdots + f_n,$$

where the $f_n$ are mutually independent, and each $f_n$ is uniformly distributed on the unit sphere $S^{d-1} \subset \mathbb{R}^d$. The underlying probability space is defined as the infinite product $X = \prod_{j=1}^{\infty} S_j$, where each $S_j = S^{d-1}$ with the usual surface measure normalized to have integral 1.

(a) If $\mu$ is the distribution measure of each $f_n$, connect $\hat{\mu}(\xi)$ with Bessel functions.

(b) What is the covariance matrix?

(c) What is the limiting distribution of $s_n(x)/n^{1/2}$?

[Hint: Show that $\hat{\mu}(\xi) = \Gamma(d/2)(\pi|\xi|)^{(2-d)/2} J_{(d-2)/2}(2\pi|\xi|)$ by using the formulas in Problem 2, Chapter 6 of Book I.]

## Chapter 6. An Introduction to Brownian Motion

**1.** The condition of continuity of Brownian paths B-3 is in effect a consequence of properties B-1 and B-2. This is implied by the following general theorem.

Suppose that for each $t \geq 0$, we are given an $L^p$ function $F_t = F_t(x)$ on the space $(X, m)$. Assume that $\|F_{t_1} - F_{t_2}\|_{L^p} \leq c|t_1 - t_2|^\alpha$, with $\alpha > 1/p$, and $1 \leq p \leq \infty$. Then there is a "corrected" $\tilde{F}_t$, so that for each $t$, $F_t = \tilde{F}_t$ (almost everywhere with respect to $m$), and so that $t \mapsto \tilde{F}_t(x)$ is continuous for all $t \geq 0$, for almost every $x \in X$. Moreover the functions $t \mapsto \tilde{F}_t(x)$ satisfy a Lipschitz condition of order $\gamma$ if $\gamma < \alpha - 1/p$.

**2.** The proof of the Donsker invariance principle follows along the same lines as the proof of Theorem 3.1. Let $f_1, \ldots, f_n, \ldots$ be a sequence of identically distributed mutually independent square integrable $\mathbb{R}^d$-valued functions on a probability space $(X, m)$, each having mean zero and the identity as its covariance matrix. Define

$$S_t^{(N)} = \frac{1}{N^{1/2}} \sum_{1 \le k \le [Nt]} f_k + \frac{(Nt - [Nt])}{N^{1/2}} f_{[Nt]+1},$$

and let $\{\mu_N\}$ be the corresponding measures on $\mathcal{P}$ induced via the measure $m$ on $X$.

(a) Instead of Lemma 3.2 use Exercise 11 to show that for $T = 1$, $\eta > 0$ and $\sigma > 0$, there exists $0 < \delta < 1$ and an integer $N_0$ so that for all $0 \le t \le 1$ one has

$$m(\{x : \sup_{0<h<\delta} |S_{t+h}^{(N)} - S_t^{(N)}| > \sigma\}) \le \delta\eta, \qquad \text{for all } N \ge N_0.$$

(b) Deduce from the above that for all $T > 0$, $\epsilon > 0$, and $\sigma > 0$ there is a $\delta > 0$ so that

$$m(\{x : \sup_{0 \le t \le T,\ 0<h<\delta} |S_{t+h}^{(N)} - S_t^{(N)}| > \sigma\}) \le \epsilon, \qquad \text{for all } N \ge 1.$$

(c) Use the inequality in (b) to show that the sequence $\{\mu_N\}$ is tight.

(d) Conclude as before that $\{\mu_N\}$ converges weakly to $W$.

**3.** There are a number of other constructions of Brownian motion besides the one given in this chapter. A particularly elegant approach is based on simple Hilbert space ideas.

On $(\Omega, P)$ consider a sequence $\{f_n\}$ of independent, identically distributed $\mathbb{R}^d$-valued functions with Gaussian distribution of mean zero and covariance matrix equal to the identity. Observe that the sequence $\{f_n\}$ is an orthonormal sequence of $L^2(\Omega, \mathbb{R}^d)$. Let $\mathcal{H}$ denote the closed subspace of $L^2(\Omega, \mathbb{R}^d)$ spanned by $\{f_n\}$.

Observe that $\mathcal{H}$ is a separable infinite dimensional Hilbert space. Hence there is a unitary correspondence $U$ between $L^2([0,\infty), dx)$ and $\mathcal{H}$. Let $B_t = U(\chi_t)$ where $\chi_t$ is the characteristic function of the interval $[0,t]$. Then, each $B_t$ can be corrected as in Problem 1, so that the process $\{B_t\}$ becomes Brownian motion. In this connection see also Exercise 9 in Chapter 5.

Note that for instance, if $B_t = \sum c_n(t) f_n$, then $B_t - B_s = \sum [c_n(t) - c_n(s)] f_n$ with $\sum |c_n(t) - c_n(s)|^2 = t - s$.

**4.*** In the previous chapter, we noted that recurrence results for the (discrete) random walks depend on the dimension $d$, and in particular, whether $d \leq 2$ or $d \geq 3$ (see Theorem 2.18 in Chapter 5 and the remark that follows it).

One can establish the following results for the (continuous) Brownian motion $B_t$ in $\mathbb{R}^d$.

(a) If $d = 1$, Brownian motion hits, almost surely, every point infinitely often, in the sense that for each $x \in \mathbb{R}$ and for any $t_0 > 0$,

$$P(\{\omega : B_t(\omega) = x \text{ for some } t \geq t_0\}) = 1.$$

Thus $B_t$ is pointwise recurrent in $\mathbb{R}$.

(b) If $d \geq 2$, then for every point $x \in \mathbb{R}^d$, Brownian motion almost surely never hits that point, that is,

$$P(\{\omega : B_t(\omega) = x \text{ for some } t > 0\}) = 0.$$

So, in this case, Brownian motion is not pointwise recurrent.

(c) However if $d = 2$, then $B_t$ is recurrent in every neighborhood of every point, that is, if $D$ is any open disc with positive radius, and $t_0 > 0$, then

$$P(\{\omega : B_t(\omega) \in D \text{ for some } t \geq t_0\}) = 1.$$

(d) Finally, when $d \geq 3$, Brownian motion is transient, that is, it escapes to infinity in the sense that

$$P(\{\omega : \lim_{t\to\infty} |B_t(\omega)| = \infty\}) = 1.$$

**5.*** The law of the iterated logarithm describes the amplitude of the oscillations of Brownian motion as $t \to \infty$ and $t \to 0$: if $B_t$ is an $\mathbb{R}$-valued Brownian motion process, then for almost all $\omega$

$$\limsup_{t\to\infty} \frac{B_t(\omega)}{\sqrt{2t \log\log t}} \to 1, \qquad \liminf_{t\to\infty} \frac{B_t(\omega)}{\sqrt{2t \log\log t}} \to -1.$$

By Exercise 13, time inversion implies that for almost all $\omega$

$$\limsup_{t\to 0} \frac{B_t(\omega)}{\sqrt{2t \log\log(1/t)}} \to 1, \qquad \liminf_{t\to 0} \frac{B_t(\omega)}{\sqrt{2t \log\log(1/t)}} \to -1.$$

**6.*** There is a converse to Theorem 6.1 when $d \geq 2$: if $u(x) \to f(y)$ as $x \to y$ with $x \in \mathcal{R}$, for each continuous function $f$, then $y$ is a regular point.

[Hint: If $y$ is not regular, then, using Problem 4* (b), show that $P(\{|B^y_{\tau_y} - y| > 0\}) = 1$, hence $P(\{|B^y_{\tau_y} - y| \geq \delta\}) > 1/2$ for some $\delta > 0$. If $S_\epsilon$ denotes the sphere centered at $y$ of radius $\epsilon < \delta$, use the strong Markov property to prove that there exists $x_\epsilon \in S_\epsilon \cap \mathcal{R}$ so that $P(\{|B^{x_\epsilon}_{\tau^{x_\epsilon}} - y| \geq \delta\}) > 1/2$. Then, considering any continuous function $0 \leq f \leq 1$ on $\mathcal{R}$ with $f(y) = 1$, and $f(z) = 0$ whenever $|z - y| \geq \delta$, leads to a contradiction.]

**7.*** A simple example of a non-regular point arises when we remove from an open ball its center, with the center then becoming a non-regular point. A more interesting example of a non-regular point is given by Lebesgue's thorn with its cusp at the origin.

Suppose $d \geq 3$, and consider the ball $B = \{x \in \mathbb{R}^d : |x| < 1\}$ from which we remove the set

$$E = \{(x_1, \ldots, x_d) \in \mathbb{R}^d : \ 0 \leq x_1 \leq 1, \ x_2^2 + \cdots + x_d^2 \leq f(x_1)\}.$$

Here $f$ is continuous and $f(x) > 0$ if $x > 0$. If $f(x)$ decreases sufficiently rapidly as $x \to 0$, then the origin is non-regular for the set $\mathcal{R} = B - E$. Clearly, $\mathcal{R}$ can be modified so that its boundary is smooth except at the origin.

## Chapter 7. A Glimpse into Several Complex Variables

**1.*** Suppose $f = f(z_1, \ldots, z_n)$ is defined in a region $\Omega \subset \mathbb{C}^n$, and for each $j$, $1 \leq j \leq n$, the function $f$ is holomorphic in $z_j$ with the other variables fixed. Then $f$ is holomorphic in $\Omega$. This was shown at the start of the chapter when $f$ is continuous, and the point of this problem is that no condition on $f$ is required besides the analyticity in each separate variable.

An important ingredient in the proof of this result is an application of the Baire category theorem.

**2.*** Assume $f$ is holomorphic in a neighborhood of the origin and $f(0) = 0$. Let $f(z) = \sum a_\alpha z^\alpha$ be the power series expansion of $f$ valid near the origin. The order of the zero (at $z = 0$) is the integer $k$ that is the smallest $|\alpha|$, for which $a_\alpha \neq 0$. Then, after a linear change of variables, we can write $f(z) = c(z)P(z)$ near the origin, where $P(z) = z_n^k + a_{k-1}(z')z_n^{k-1} + \cdots + a_0(z')$ with $(z', z_n)$, and $c(z) \neq 0$ while $a_{k-1}(0) = \cdots = a_0(0) = 0$. This result is the **Weierstrass preparation** theorem.

[Hint: Assume that our coordinate system $(z', z_n) \in \mathbb{C}^{n-1} \times \mathbb{C}$ is such that $f(0, z_n) = z_n^k$. Then by Rouché's theorem we can choose $\epsilon, r > 0$, so that $z_k \to f(z', z_k)$ has $k$ zeroes inside the disc $|z_k| \leq r$, but is non-vanishing on the boundary, for all $|z'| < \epsilon$. Let $\gamma_1(z')$, $\gamma_2(z')$, $\ldots$ , $\gamma_k(z')$ be an arbitrary ordering of these zeroes. Then the symmetric functions $\sigma_1(z') = \sum_{\ell=1}^k \gamma_\ell(z')$, $\sigma_2(z') = \sum_{m<\ell} \gamma_\ell(z')\gamma_m(z'), \ldots$ , are holomorphic in $z'$, for $|z'| < \epsilon$. This follows since the sums $s_m(z') = \sum_{\ell=1}^k (\gamma_\ell(z'))^m$, $1 \leq m \leq k$, have this property because they are given by the formula

$$s_m(z') = \frac{1}{2\pi i} \int_{|w|=r} w^m \frac{(\partial f/\partial w)(z', w)}{f(z', w)}\, dw.$$

Now we need only take $a_{k-j}(z') = (-1)^j \sigma_j(z')$, and the result holds for $P(z) = z_n^k + a_{k-1}(z')z^{k-1} + \cdots + a_0(z')$.]

**3.*** The original proof of Theorem 4.1 represented $F$ in terms of $F_0$ by Green's theorem via the "Bochner-Martinelli integral." The result then held for $F_0$ merely of class $C^1$.

**4.*** We are concerned with the problem

$$\overline{\partial} u = f, \qquad \text{on } \Omega, \tag{42}$$

where $\Omega$ is a bounded region in $\mathbb{C}^n$ with $C^\infty$ boundary and $f$ is given in $\Omega$ with $\overline{\partial} f = 0$ there.

(a) If $\Omega$ is pseudo-convex, and $f \in C^\infty(\overline{\Omega})$ then there is $u \in C^\infty(\overline{\Omega})$ that solves (42).

(b) The "normal" solution (if it exists) is defined as the (unique) solution $u$ in $L^2(\Omega)$ for which

$$\int_\Omega u\overline{F}\, dm(z) = 0$$

for all $F$ that are holomorphic in $\Omega$ and are in $L^2$ there. For $\Omega$ that are strongly pseudo-convex (and many other classes of $\Omega$) whenever $f \in C^\infty(\overline{\Omega})$, the normal solution $u$ also belongs to $C^\infty(\overline{\Omega})$. This results from the study of the "$\overline{\partial}$-Neumann problem."

**5.* Domains of holomorphy.** A domain of holomorphy is a region $\Omega$ with the property that there exists a holomorphic function $F$ so that for every $z^0 \in \partial\Omega$, the function $F$ cannot be continued into some ball centered at $z^0$. If $\Omega$ is a domain of holomorphy and has boundary of class $C^2$, then $\Omega$ is pseudo-convex, by Theorem 7.5. Conversely, it can be shown that if $\Omega$ is pseudo-convex, then it is a domain of holomorphy.

**6.*** The converse to Theorem 8.7 holds. If $f$ is a distribution with compact support so that $C(f)(z)$ is analytic near $z = 0$ then $\overline{L}(U) = f$ is locally solvable near the origin.

This is proved by finding a kernel $K$ so that the convolution operator $T(f) = f * K$ on the Heisenberg group is a relative inverse to $\overline{L}$ in the sense that $\overline{L}T(f) = f - C(f)$. Then write $f = f - C(f) + C(f) = f_1 + f_2$, with $f_1 = f - C(f)$ and $f_2 = C(f)$. We can solve $L(U_1) = f_1$ by what has just been asserted, and we can solve $L(Uu_2) = f_2$ locally by the Cauchy-Kowaleski theorem, since $f_2$ is real-analytic at the origin.

## Chapter 8. Oscillatory Integrals in Fourier Analysis

**1.*** Suppose $M$ is a local hypersurface in $\mathbb{R}^d$. In a neighborhood of a point $x_0 \in M$ one can choose a smooth vector field $\nu$, defined in this neighborhood restricted to $M$, so that $\nu(x)$ is a unit normal vector of $M$ at each $x \in M$. (There are two choices of this vector field, determined up to a sign.) The map $x \mapsto \nu(x)$ from $M$ to $S^{d-1}$ (with $S^{d-1}$ the unit sphere in $\mathbb{R}^d$) is called the **Gauss map**.

One can prove that the Gauss curvature of $M$ near $x_0$ is non-vanishing if and only if the Gauss map is a diffeomorphism near $x_0$. Moreover, if $d\sigma_M$ and $d\sigma_{S^{d-1}}$ are the induced Lebesgue measures of $M$ and $S^{d-1}$, and $(d\sigma_{S^{d-1}})^*$ the pull-back of $d\sigma_{S^{d-1}}$ to $M$ defined by

$$\int_M f\,(d\sigma_{S^{d-1}})^* = \int_{S^{d-1}} f(\nu^{-1}(x))\,d\sigma_{S^{d-1}}(x),$$

then $K d\sigma_M = (d\sigma_{S^{d-1}})^*$, with $K$ the absolute value of the Gauss curvature.

**2.* The spherical maximal function.** Define

$$A_t(f)(x) = \frac{1}{\sigma(S^d)} \int_{S^d} f(x - ty)\, d\sigma(y)$$

for each $t \neq 0$, and $A^*(f)(x) = \sup_{t \neq 0} |A_t(f)(x)|$. Then

$$\|A^*(f)\|_{L^p} \leq c_p \|f\|_{L^p}, \qquad \text{if } p > d/(d-1) \text{ and } d \geq 2.$$

As a result, if $f \in L^p$, $p > d/(d-1)$, then $\lim_{t \to 0} A_t(f)(x) = f(x)$ a.e. Simple examples show that this fails if $p \leq d/(d-1)$.

A hint that there may be estimates for $\sup_t |A_t(f)|$ (and in particular that the result holds for $p = 2$ and $d \geq 3$) is the following simple observation for $d \geq 3$:

$$\| \sup_{1 \leq t \leq 2} |A_t(f)| \, \|_{L^2} \leq c \|f\|_{L^2}.$$

To establish this, one notes that

$$\int_1^2 \left| \frac{\partial A_s(f)(x)}{\partial s} \right|^2 dx\, ds \leq c' \|f\|_{L^2}^2$$

by using Theorem 3.1. However $\sup_{1 \leq t \leq 2} |A_t(f)| \leq \int_1^2 \left| \frac{\partial A_s(f)}{\partial s} \right| ds + |A_1(f)(x)|$, hence the assertion follows by using Schwartz's inequality.

Refinements of this argument prove the result for $\sup_{t>0} |A_t(f)|$, $p = 2$ and $d \geq 3$, and then also for $p > d/(d-1)$. Further ideas are needed for the case $d = 2$.

**3.*** There is a variant of Problem 2 that applies to the wave equation.

Suppose $u$ solves $\triangle_x u = \frac{\partial^2 u}{\partial t^2}$ for $(x,t) \in \mathbb{R}^d \times \mathbb{R}$, with $u(x,0) = 0$, and $\frac{\partial u}{\partial t}(x,0) = f(x)$. If $f \in L^2$ we observe that $\frac{u(x,t)}{t} \to f(x)$ in the $L^2(\mathbb{R}^d)$ norm as $t \to 0$. One can show that $\lim_{t\to 0} \frac{u(x,t)}{t}$ exists and equals $f(x)$ a.e. if $f \in L^p$, $p > 2d/(d+1)$.

**4.*** The restriction phenomenon (inequality (31)) is valid in $\mathbb{R}^2$, for the full range $1 \leq p < 4/3$.

[Hint: One may dualize the assertion as in the proof of Theorem 5.2. Consider the operator $\mathcal{R}^*$ defined by

$$\mathcal{R}^*(F)(x) = \int_M e^{2\pi i x\cdot\xi} F(\xi)\, d\mu(\xi).$$

The desired result then becomes the inequality

$$\|\mathcal{R}^*(F)\|_{L^q(\mathbb{R}^2)} \leq A\|F\|_{L^p(d\mu)}$$

when $q = 3p'$ and $1 \leq p < 4$. Now the key point is that if we consider the singular measure $d\nu = Fd\mu$, then the convolution $\nu * \nu$ is actually an absolutely continuous measure $f\,dx$ with density $f$, a locally integrable function on $\mathbb{R}^2$. This fact reflects the assumed curvature of $M$. Indeed, it can be shown that $f \in L^p(\mathbb{R}^2)$, with $\frac{3}{r} = \frac{2}{p} + 1$, whenever $F \in L^p(d\mu)$ and $1 \leq p \leq 4$, and $\|f\|_{L^r(\mathbb{R}^2)} \leq c\|F\|^2_{L^p(d\mu)}$ and $1 \leq p < 4$. Now if this is so, then

$$\mathcal{R}^*(F)^2 = (\hat{\nu}(-x))^2 = (\nu * \nu)^\wedge(-x) = \hat{f}(x),$$

and by the Hausdorff-Young inequality,

$$\|\mathcal{R}^*(F)\|^2_{L^{2r'}} \leq \|(\mathcal{R}^*(F))^2\|_{L^{r'}} = \|\hat{f}\|_{L^{r'}} \leq c\|F\|^2_{L^p},$$

and this proves the assertion since $2r' = 3p'$.]

**5.**$^*$ An analog of Theorem 6.3 for the wave equation is as follows. Let $u(x,t)$ be the solution of the wave equation $\frac{\partial^2 u}{\partial t^2} = \triangle u$ for $(x,t) \in \mathbb{R}^d \times \mathbb{R}$, with initial data

$$\begin{cases} u(x,0) &= 0 \\ \frac{\partial u}{\partial t}(x,0) &= f(x). \end{cases}$$

Then $\|u(x,t)\|_{L^q(\mathbb{R}^d \times \mathbb{R})} \leq c\|f\|_{L^2(\mathbb{R}^d)}$ if $q = \frac{2d+2}{d-2}$ and $d \geq 3$.

**6.*** The following further results are known about $E(R) = N(R) - \pi R^2$, the error term appearing in Theorem 8.3.

(a) The Hardy series $R\sum_{k=1}^{\infty} \frac{r_2(k)}{k^{1/2}} J_1(2\pi k^{1/2} R)$ converges for each $R \geq 0$, and its sum equals $E(R)$ whenever $R \neq k^{1/2}$, for any positive integer $k$.

(b) The error $E(R)$ is on the average a multiple of $R^{1/2}$ in the sense that

$$\int_0^r E(R)^2 R\, dR = cr^3 + O(r^{2+\epsilon}),$$

for some $c > 0$ and every $\epsilon > 0$.

(c) However, $E(R)$ is not exactly $O(R^{1/2})$ since

$$\limsup_{R\to\infty} \frac{|E(R)|}{R^{1/2}} = \infty.$$

(d) It has been proved that $E(R) = O(R^{\alpha+\epsilon})$, for certain $\alpha$, $1/2 < \alpha < 2/3$. A relatively recent result of this kind is for $\alpha = 131/208$.

**7.**$^*$ The oscillatory integral $\mathfrak{J}(\lambda)$ can be identified in terms of Bessel functions of the second and third kind. One has that

$$\mathfrak{J}(\lambda) = 4K_0(2\lambda) - 2\pi Y_0(2\lambda),$$

where $Y_m$ and $K_m$ are respectively Neumann and Macdonald functions.

**8.*** Consider the error term in the divisor problem

$$\Delta(\mu) = \sum_{k=1}^{\mu} d(k) - \mu \log \mu - (2\gamma - 1)\mu - 1/4.$$

It is given for $\mu$ not an integer by the convergent series

$$\frac{-2}{\pi}\mu^{1/2} \sum_{k=1}^{\infty} \frac{d(k)}{k^{1/2}} \left[ K_1(4\pi k^{1/2}\mu^{1/2}) + \frac{\pi}{2} Y_1(4\pi k^{1/2}\mu^{1/2}) \right].$$

For $\Delta$ there are estimates, analogous to those for $E$ in Problem 6, with $\Delta(\mu) = O(\mu^{\beta+\epsilon})$ and $\beta = \alpha/2$.